Optimizing Traffic Signal Scheduling:

A Game Theory Approach to Reducing Wait Times

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Table of content

Literature Review

Sr. No	Year	Paper Title	Methods	Limitations
1	2020	Game-Theoretic Traffic Signal Control Framework	Non-cooperative game model where intersections act as independent agents optimising signal times.	Limited scalability; agents focus on local optimisation rather than network-wide fairness.
2	2021	Multi-Agent Reinforcement Learning for Traffic Signal Optimization	Combined reinforcement learning with Game Theory to coordinate multiple intersections.	High computational cost; requires extensive training data for stable performance.
3	2022	Cooperative Game Theory in Multi-Intersection Signal Control	Applied cooperative Game Theory where agents share traffic data for collective optimisation.	Increased communication overhead; assumes ideal coordination among all intersections.
4	2023	Hybrid Optimisation and Game-Theoretic Scheduling	Integrated classical optimisation algorithms (GA, PSO) with game-theoretic decision models.	Sensitive to hyperparameter tuning; less effective under sudden traffic variations.
5	2024	Intelligent Traffic Management: A Game-Theory Perspective	Surveyed decentralized optimization and multi-agent coordination for smart cities.	Computational intensity; limited focus on fairness and low-cost implementation.

Problem Statement

"Conventional traffic signals use fixed-time cycles, which are inefficient and cannot adapt to real-time traffic fluctuations, leading to unnecessary congestion and long delays."

Use-Case Diagram

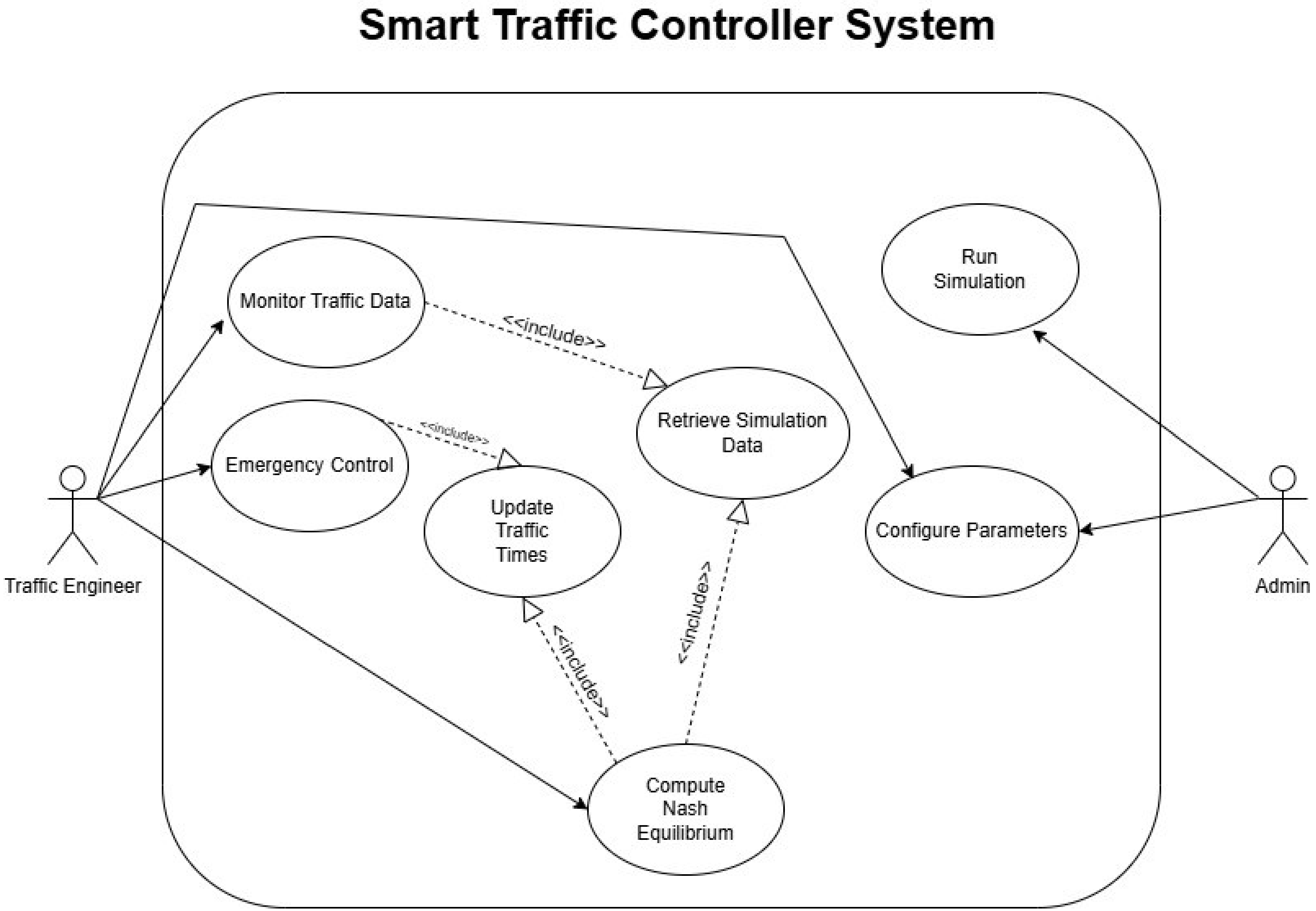


Fig 1: Use case Diagram

Workflow Diagram

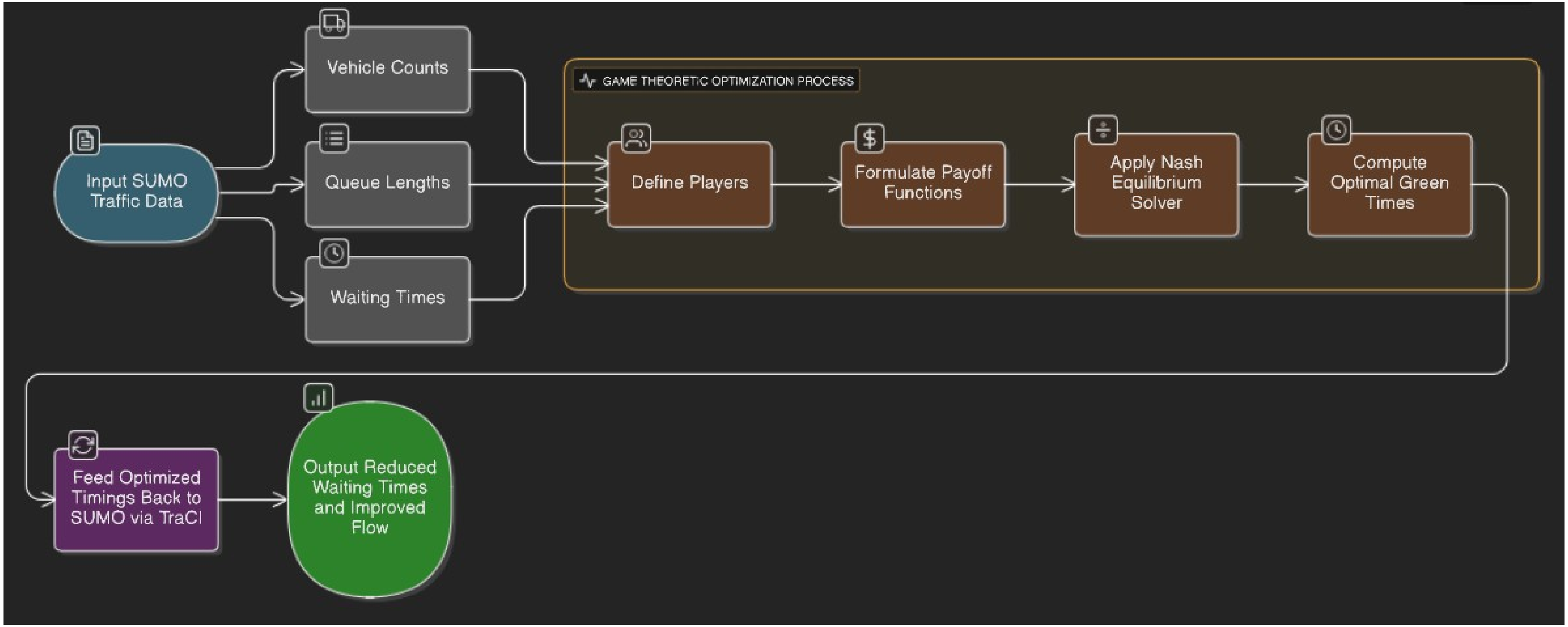


Fig 2: Workflow Diagram

Architecture Diagram

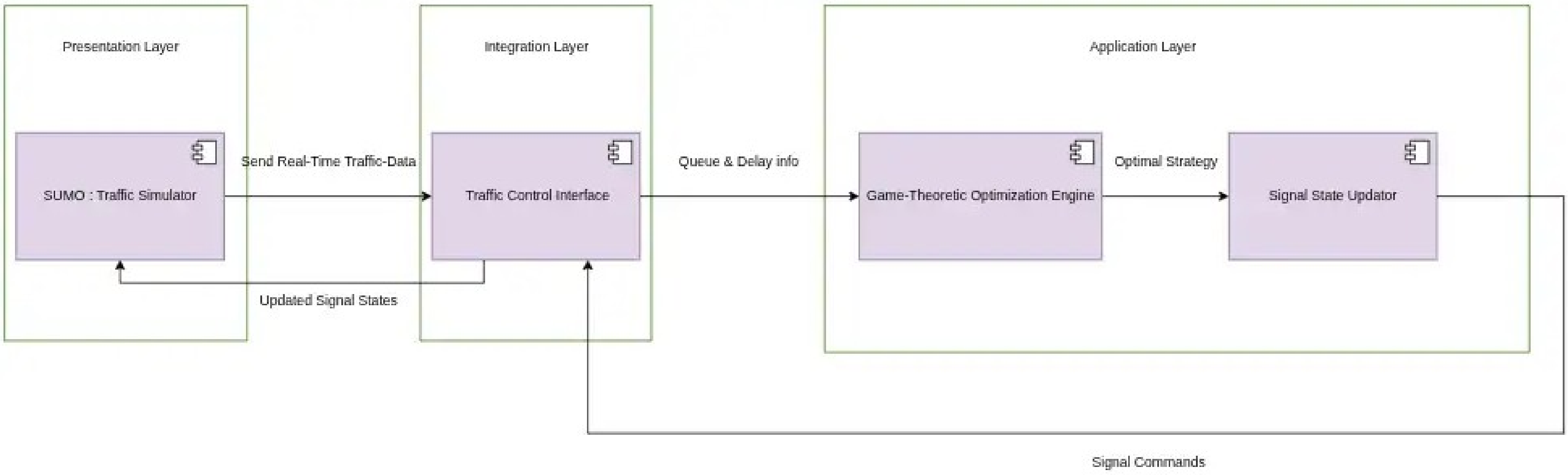
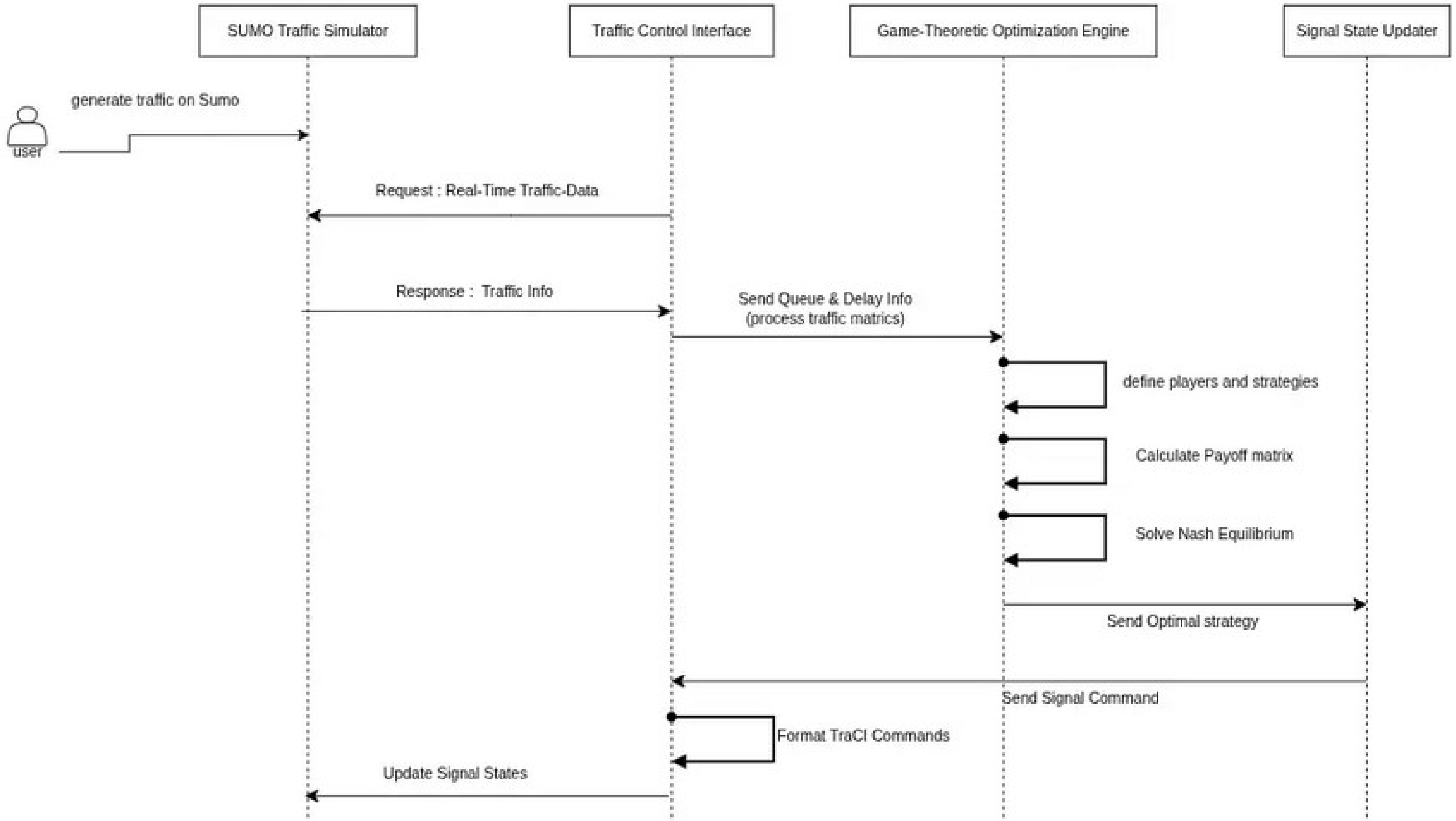


Fig 3: Architecture Diagram

Activity Diagram



Implementation Methodology

INPUT:
- T_{cycle}: Total signal cycle time (e.g., 100 seconds)
- g_{min}, g_{max}: Minimum and maximum green time bounds
- τ_{yellow}, τ_{all_red}: Clearance times (yellow + all-red)
- α, β, γ: Weight coefficients for payoff function
- edge_ids: SUMO edge identifiers [North, East, South, West]
- ε: Convergence threshold
- η: Learning rate
- max_iterations: Maximum number of iterations

OUTPUT:
- g_{NS}^*, g_{EW}^*: Optimal green times for North-South and East-West phases
- performance_metrics: (avg_wait, total_delay, throughput, convergence_status)

Implementation Methodology

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GAME FORMULATION:
    Players: 4 approaches {North, East, South, West}

    Strategy Space:
        S_i = {g_i | g_min ≤ g_i ≤ g_max}

    Payoff Function:
        U_i(g) = -(α·W_i + β·D_i + γ·Q_residual)

        Where:
            W_i = Q_i × (T_cycle - g_i)           // Waiting cost
            D_i = Q_i² / (2 × s_i × g_i)          // Delay cost (Webster's formula)
            Q_residual = max(0, Q_i - s_i × g_i) // Unserved vehicles
            s_i = saturation_flow × 2             // For 2 lanes per approach

    Constraints:
        g_NS + g_EW + 2×(τ_yellow + τ_all_red) = T_cycle
        g_NS = g_North = g_South                  // Paired movements
        g_EW = g_East = g_West                    // Paired movements
```

Project Planning

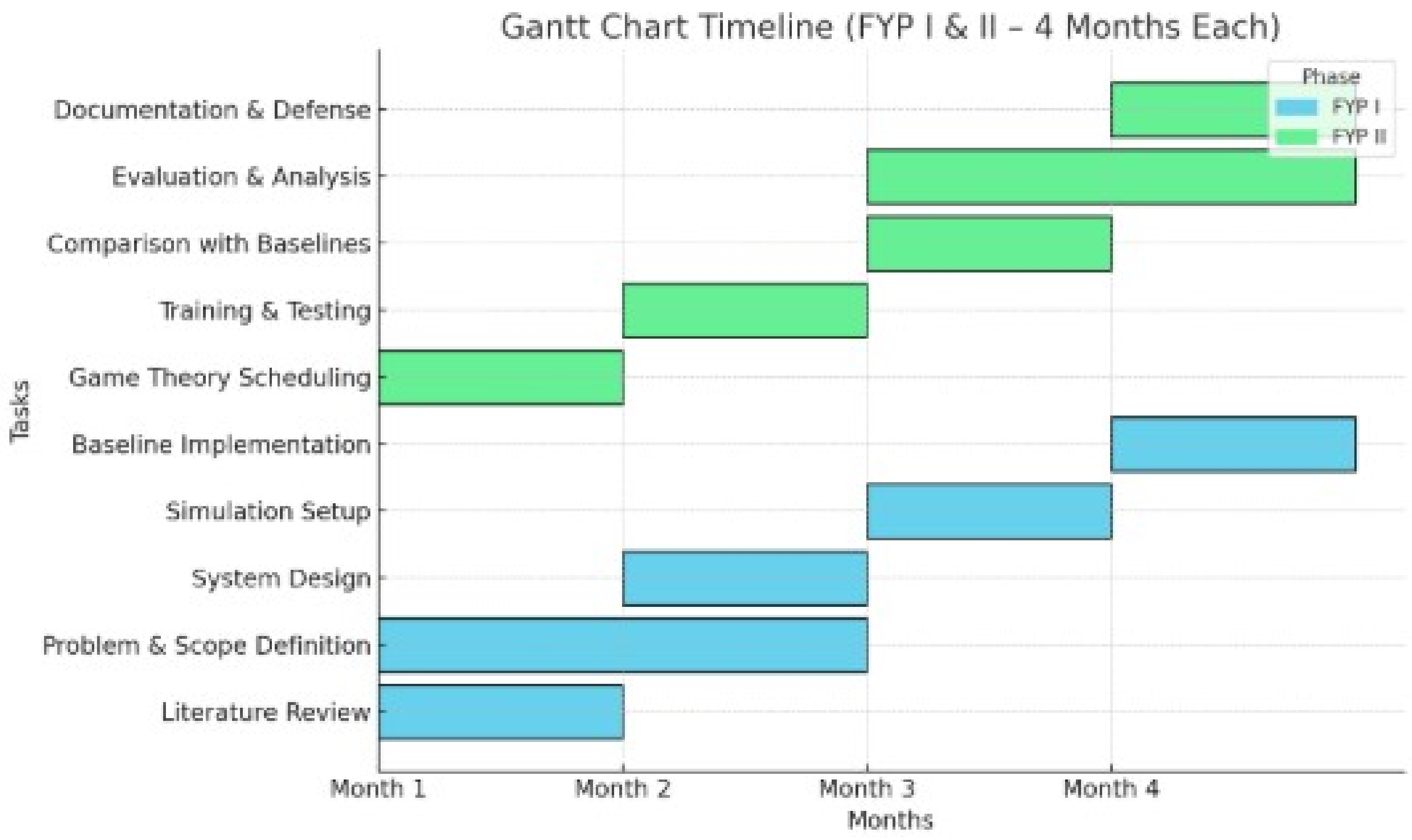

Fig 5: Gantt Chart

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HAVE A GOOD DAY!